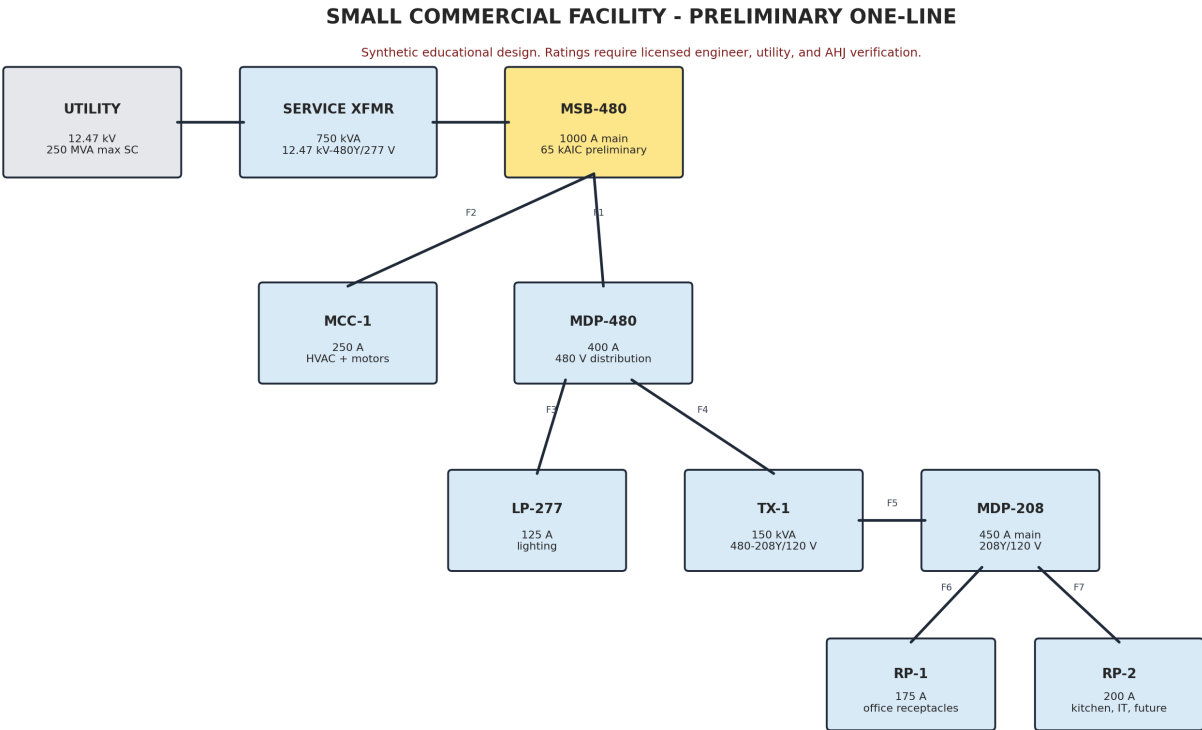


Small Commercial Facility Power Distribution Design and Analysis

Engineering report

Synthetic educational design - not for construction
Prepared by Giorgi Gogitidze
September 2026



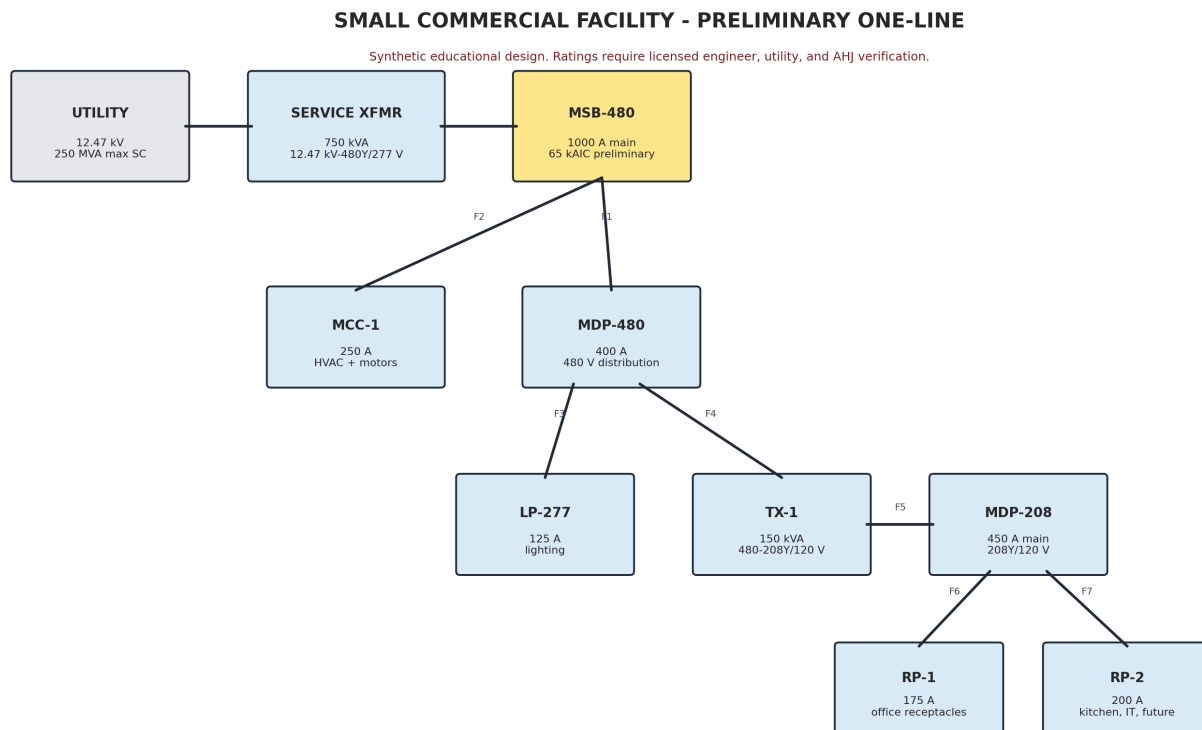
This report documents an executable portfolio project. No real facility, utility account, field survey, proprietary-software result, or professional seal is represented.

1. Executive Summary

A radial 12.47 kV to 480Y/277 V to 208Y/120 V system was designed for a fictional 35,000 ft² office/warehouse. The corrected base model supplies approximately 331 kW and 142 kvar. Minimum voltage is 0.952 pu, maximum feeder loading is 69.5%, and both the service and 150 kVA transformers operate near 48%.

The peak, future-expansion, and low-source-voltage cases fall below the project 0.95 pu planning criterion. Voltage regulation, rather than transformer kVA, is the controlling future issue. Maximum modeled three-phase fault current is about 16.2 kA at MSB-480. SLG values remain explicit approximations.

2. System Architecture



The 480 V system supplies HVAC, motors, heating, process load, and 277 V lighting. A 150 kVA delta-grounded-wye transformer establishes a separately derived 208Y/120 V system for receptacle, IT, kitchen, and future tenant loads.

3. Design Basis and Assumptions

Utility source: 12.47 kV, 60 Hz, 250 MVA maximum and 150 MVA minimum short-circuit strength. Service transformer: 750 kVA, 5.75% impedance. TX-1: 150 kVA, 480-208Y/120 V, 5.75% impedance. Conductors are assumed copper THHN/THWN-2 using preliminary 75 C ampacity values.

Demand factors, load power factors, cable impedances, equipment ratings, and grounding data are synthetic. The locally adopted NEC edition, amendments, utility requirements, actual terminal temperatures, derating, and manufacturer data must be verified for any real project.

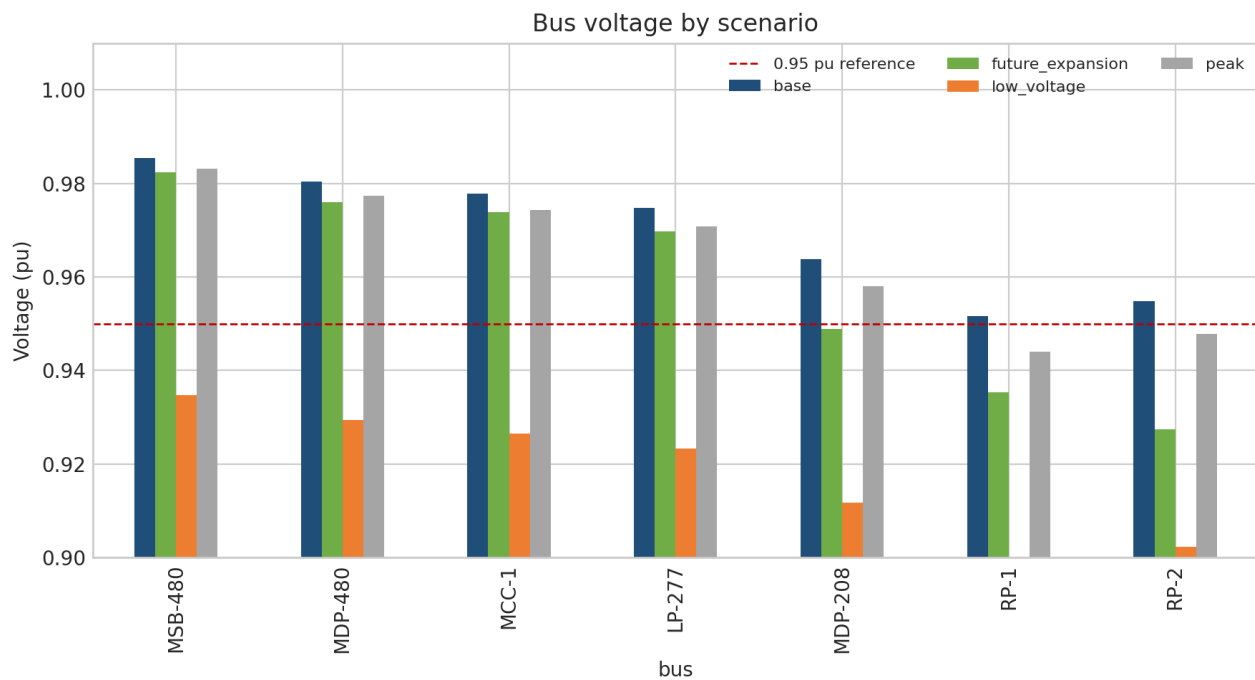
4. Load and Sizing Method

Demand real power is connected kW times demand factor. Reactive power is $Q = P \tan(\cos^{-1}(PF))$. Balanced three-phase current is $I = P / (\sqrt{3} V PF)$; line-to-neutral current is $I = P / (V PF)$. Equipment and conductors were selected provisionally and then tested in the executable model.

The first model run found that a 225 A MDP feeder had omitted downstream demand and exceeded 100% loading. The corrected design uses two parallel 3/0 Cu sets on a 400 A breaker. A phase-allocation check also caused the kitchen circuit to move from Phase A to Phase C.

5. Scenario Results

Scenario	Source kW	Min pu	Max line %	TX-1 %	Service TX %
base	331.4	0.952	69.5	48.1	48.1
peak	381.7	0.944	80.5	55.6	55.4
future_expansion	397.4	0.927	89.9	76.7	57.8
low_voltage	331.7	0.899	73.6	50.8	50.7



The base case is thermally acceptable and just above 0.95 pu. Peak and future cases require mitigation such as lower-impedance/shorter 208 V feeders, transformer tap review, a different future-load voltage, or power-factor correction after harmonic assessment.

6. Voltage-Drop Checks

Circuit	Current A	PF	Drop V	Drop %
F1	249.4	0.96	2.32	0.48
F2	169.8	0.88	3.62	0.75
F3	82.3	0.95	2.67	0.56
F4	81.3	0.96	1.01	0.21
F5	187.6	0.96	0.36	0.17
F6	115.7	0.95	2.40	1.16
F7	72.1	0.97	1.77	0.85
BC-1	16.0	0.95	4.47	3.73

The representative 120 V branch was upsized from #12 to #10 Cu after the initial calculation yielded about 5.9% branch drop. The revised branch drop is about 3.7%, and the combined feeder-plus-branch path is about 4.9%.

7. Fault Current and Protection

Bus	V	3ph kA	SLG est. kA
MSB-480	480	16.16	9.70
MDP-480	480	13.27	7.96
MCC-1	480	10.27	6.16
LP-277	480	7.96	4.78
MDP-208	208	5.97	3.58
RP-1	208	4.16	2.49
RP-2	208	3.79	2.27

Three-phase values use pandapower IEC 60909 positive-sequence calculations. The transformer-only infinite-bus check for TX-1 is 7.24 kA; the network model gives 5.97 kA after upstream impedance. SLG values assume $Z_2=Z_1$ and $Z_0=3Z_1$ and must not be used for final equipment selection.

No manufacturer time-current curves were modeled. Preliminary AIC selections exceed modeled duties, but series ratings, exact catalog numbers, transformer inrush/damage, motor starting, selective coordination, and IEEE 1584 arc-flash clearing times remain open.

8. Validation

Four unit tests verify three-phase current, transformer terminal fault current, voltage-drop behavior, and phase balance. All four load-flow scenarios converge. Independent hand checks agree with model direction and magnitude. The project regenerates source-controlled CSV, PNG, SVG, XLSX, and PDF artifacts.

9. Limitations

The model is balanced and positive-sequence. It omits unbalanced neutral current, harmonics, motor starting, generator/PV contribution, detailed grounding paths, temperature/raceway derating, equipment-enclosure effects, protection curves, arc flash, cost estimation, and construction drawings. AutoCAD, Revit, and ETAP verification was not performed.

10. Recommendations

Obtain utility sequence data and transformer test reports; replace generic cable impedances; confirm actual routes and terminal ratings; evaluate transformer taps and 208 V feeder upgrades; build an unbalanced four-wire model; import exact protection curves; run motor-starting, coordination, and arc-flash studies; and obtain licensed professional engineer and AHJ review.

References

NFPA 70, National Electrical Code, adopted edition and local amendments to be confirmed. IEEE 3002.2-2018, load-flow studies for industrial and commercial systems. IEEE 3002.3-2018, short-circuit studies for industrial and commercial systems. Pandapower 3.2.1 documentation and IEC 60909 implementation.